

UTKARSH CHAVAN

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PROFESSIONAL SUMMARY

B.Tech Information Technology student (minor in Data Science) with hands-on experience in full-stack web development, AI/ML, computer vision, and Android applications through hackathons and independent projects. Interested in building practical software and data-driven solutions to real-world problems.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML, CSS, Kotlin

Concepts: Data Structures & Algorithms, OOP, REST APIs, Machine Learning, Computer Vision, Data Processing, Agile Methodology, Problem Solving

Frameworks & Libraries: Flask, FastAPI, Angular, Node.js, Express.js, OpenCV, YOLOv8, Scikit-learn, XGBoost, Pandas, NumPy

Databases: MySQL, PostgreSQL, SQLite

Cloud & DevOps: AWS S3, Docker, Kubernetes, CI/CD Pipelines, Firebase, Git, GitHub

Tools & Platforms: VS Code, Android Studio, Google Colab, Postman, Swagger

EDUCATION

B.Tech in Information Technology (Minor: Data Science)

2023 – 2027

MGM University – Institute of Information and Communication Technology (IICT), Chh. Sambhajinagar

SGPA: 8.5

Higher Secondary Certificate (HSC)

2023

Defence Career Institute — Score: 77.64%

Secondary School Certificate (SSC)

2021

Aurangabad Police Public School — Score: 91.60%

PROJECTS

Smart City AI Road Hazard Detection Platform

- Developed an AI-powered real-time road hazard detection platform using Python (FastAPI), SQLite, YOLOv8, and OpenCV.
- Engineered a WebSocket-based real-time pipeline to detect and broadcast hazards including potholes, accidents, and obstacles.
- Built an interactive monitoring dashboard using Leaflet.js with live alerts, hazard analytics, and map-based visualization.

MediCare – AI-Powered Health Diagnostic Web Application

- Developed a symptom-based diagnostic system using Python (Flask) and a Random Forest classifier during a 24-hour hackathon.
- Implemented auto-suggest symptom input, intensity scoring, and fuzzy matching to improve user interaction.
- Delivered personalized health guidance through a responsive frontend built with HTML, CSS, and JavaScript.

Zero UI – Sensor-Based Emergency Alert System

Finalist, Zero UI Challenge

- Designing a sensor-driven emergency detection system on Android (Kotlin, Jetpack Compose) using accelerometer and gyroscope data to detect abnormal motion and free-fall events.
- Built an automatic detection flow (impact detection → post-impact verification → user confirmation → emergency response) alongside a manual panic-mode trigger.
- Integrated GPS-based location tracking and continuous background monitoring via an Android Foreground Service; backend (FastAPI) and responder dashboard planned for future development.

Exoplanet Detection System

In Progress

- Building an end-to-end light-curve analysis and AI-assisted classification pipeline to identify exoplanet candidates from NASA TESS light-curve data using Python, Lightcurve, and Astropy.
- Applying the Transit Least Squares (TLS) algorithm to detect candidate transit events after preprocessing and noise/trend removal.
- Extracting transit and astrophysical features for candidate validation, with a Scikit-learn/XGBoost classifier planned to rank candidates by confidence.

NGO-Volunteer Connect Platform

In Progress – Capstone Project

- Developing a full-stack platform (Angular, Node.js, Express.js, PostgreSQL) connecting individual volunteers with NGOs, featuring a core “Need Matching” system that links NGO-posted needs directly with capable volunteers.
- Implementing JWT-based authentication and documented REST APIs (Swagger), with role-based access for individuals, NGOs, and admins.

CERTIFICATIONS & TRAINING

- Full Stack Development Program, Sure Trust — Ongoing, Remote (Since May 2026)
- Data Lake & Cloud Fundamentals Workshop — AWS S3, Docker, Kubernetes, CI/CD Pipelines, Git/GitHub
- Fundamentals of Machine Learning, Saylor Academy
- UI/UX Design Workshop (S8), Rablo.in via Unstop
- Generative AI Powered Data Analytics Virtual Internship, Tata Group via Forage

ACHIEVEMENTS & LEADERSHIP

- Finalist – Hack the Gap, 24-hour hackathon conducted by IEEE MG MU.
- Finalist – Zero UI Challenge, conducted by Cummins College of Engineering for Women, Pune.
- Participated in Smart India Hackathon, Google Solution Challenge, and Hack2Skill Challenge.
- Logistics and Hospitality Head for college cultural events — managed planning, coordination, and execution.